A Minimal Research Project Template

An Illustrative Example in Python

Course Template

October 19, 2026

Abstract

This paper demonstrates a minimal reproducible Python workflow for class assignments. The project uses a tiny `mtcars` example, but preserves the basic logic of a larger empirical workflow: data are pulled, prepared, analyzed, serialized to disk, and then imported into the paper as prepared results.

1 Introduction

This repository is the Python version of a barebones project template for class assignments. It keeps the moving parts of a larger workflow visible while staying small enough to understand in one sitting. Data move through a pull step, a preparation step, an analysis step, and finally into a paper that reads saved results rather than re-running the analytical pipeline. The paper structure is intentionally compact, but it already includes standard scholarly features such as citations and a bibliography (Wickham 2014).

2 Data

The example uses the built-in `mtcars` dataset that ships with `plotnine`. In this template, `code/python/pull_data.py` writes a raw object to `data/pulled/`, `code/python/prep_data.py` creates a prepared analysis dataset in `data/generated/`, and `code/python/run_analysis.py` reads that prepared dataset to create the final analytical objects stored in `output/`.

The prepared sample contains 32 observations. Average fuel efficiency is 20.1 miles per gallon, average horsepower is 146.7, and the heaviest vehicle in the sample is 15.

3 Analysis

The analysis step estimates a simple linear model of miles per gallon as a function of vehicle weight and transmission type. The goal is not substantive complexity. The goal is to show how a project can separate raw data handling, preparation, analysis, and reporting into distinct scripts that produce durable artifacts.

4 Results

[Table 1 about here.]

[Figure 1 about here.]

The fitted model implies a -5.35 mpg change for an additional 1,000 pounds of vehicle weight. The coefficient on manual transmission is -0.02 mpg. The prepared sample suggests a clear

negative relationship between vehicle weight and fuel efficiency, while the manual cars in this small dataset have somewhat higher fuel efficiency after controlling for weight.

5 Conclusion

This repository is intentionally small, but it is meant to be built on. Students can replace the toy data step, extend the preparation logic, add richer analysis, and continue writing in the same paper structure without changing the overall workflow shape.

A visible placeholder remains here for the in-class GitHub edit if needed: `YOUR NAME - summary`.

References

Wickham, Hadley. 2014. "Tidy Data." *Journal of Statistical Software* 59 (10): 1–23. https://doi.org/10.18637/jss.v059.i10.

Figure 1: Fuel efficiency declines with vehicle weight in the prepared sample, with manual cars concentrated among the lighter and more efficient models.

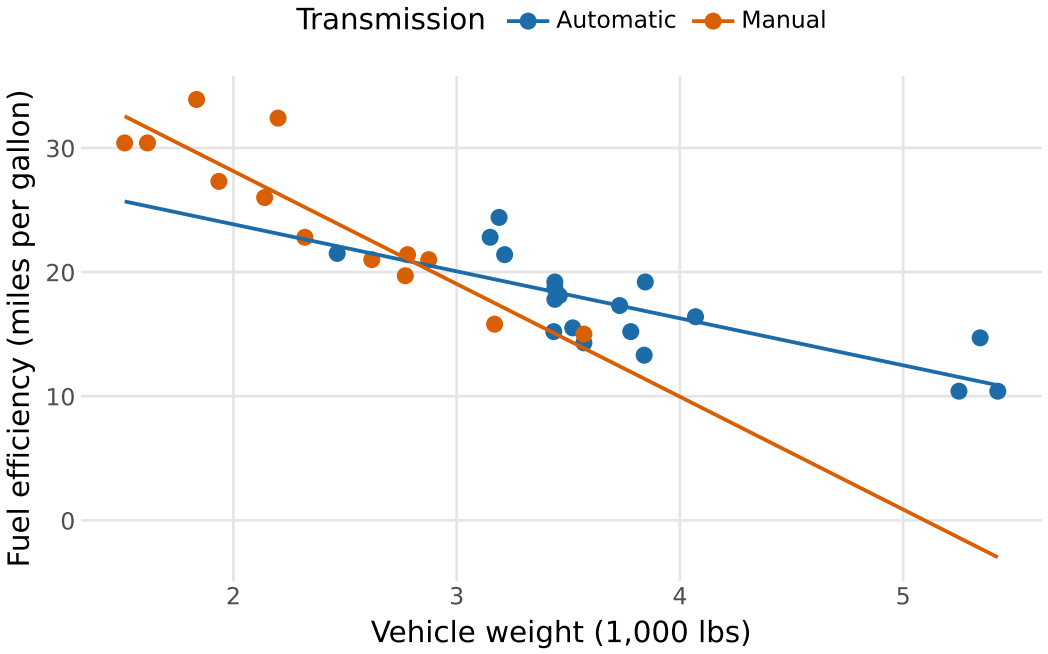

Table 1: Descriptive statistics by transmission show that manual cars in the prepared sample are lighter and more fuel efficient on average.

Transmission	Cars	Mean fuel efficiency (mpg)	Mean horsepower	Mean weight (1,000 lbs)
Automatic	19	17.1	160.3	3.8
Manual	13	24.4	126.8	2.4

Note: This table summarizes the prepared mtcars sample by transmission type. Fuel efficiency is measured in miles per gallon and weight is measured in 1,000 pounds.